balance) while inflation declines (by ¼ percentage points for each percentage point decrease in GDP growth). Debt rises to 45 percent of GDP in 2020 and gradually rises to 54 percent in the medium term. Gross public financing needs peak in 2020 before declining to about 14 percent of GDP over the medium term.

- **Combined macro-fiscal shock.** A combined macro-fiscal shock incorporates the largest effect of the individual macro-fiscal shocks (i.e., excluding the contingent liability shock) on all relevant variables (real GDP growth, inflation, primary balance, exchange rate, and the interest rate). Public debt would reach 62 percent of GDP over the medium term. Gross financing needs would rise to 18 percent over the medium term.
- **Combined macro-fiscal-contingent liability shock.** This extreme combined shock incorporates the largest effect of the above shocks, including the contingent liability shock, on all relevant variables. Public debt would breach the 70 percent of GDP benchmark by 2024 under this shock scenario. Gross financing needs would also increase to above 20 percent of GDP.

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Figure 1. Turkey Public Sector Debt Sustainability Analysis (DSA)—Baseline Scenario

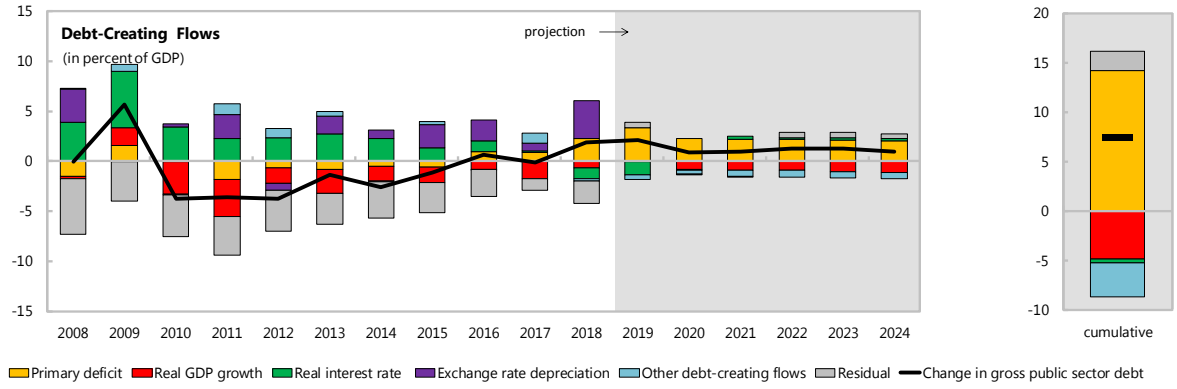
(in percent of GDP unless otherwise indicated)

Debt, Economic and Market Indicators [1/]

	Actual			Projections						As of November 13, 2019		
	2008-2016 2/	2017	2018	2019	2020	2021	2022	2023	2024	Sovereign Spreads		
Nominal gross public debt	34.2	28.2	30.1	32.2	33.1	34.1	35.4	36.6	37.6	EMBIG (bp) 3/		415
Public gross financing needs	9.9	5.0	6.5	7.5	9.2	10.2	10.4	10.6	10.0	5Y CDS (bp)		316
Real GDP growth (in percent)	4.8	7.5	2.8	0.2	3.0	3.0	3.0	3.5	3.5	Ratings	Foreign	Local
Inflation (GDP deflator, in percent)	7.7	11.0	16.4	14.4	11.8	11.6	10.9	10.8	11.0	Moody's	B1	B1
Nominal GDP growth (in percent)	12.9	19.2	19.7	14.7	15.1	15.0	14.3	14.7	14.9	S&Ps	B+	BB-
Effective interest rate (in percent) [4/]	10.8	8.1	9.0	9.6	11.9	13.1	12.0	11.9	12.0	Fitch	BB-	BB-

Contribution to Changes in Public Debt

	Actual			Projections							
	2008-2016 2/	2017	2018	2019	2020	2021	2022	2023	2024	cumulative	debt-stabilizing primary balance [9/]
Change in gross public sector debt	-1.1	-0.1	1.9	2.1	0.9	1.0	1.3	1.2	1.0	7.5	
Identified debt-creating flows	2.7	1.0	4.1	1.6	1.0	1.0	0.8	0.7	0.5	5.5	
Primary deficit	-0.4	0.9	2.3	3.4	2.3	2.2	2.2	2.1	2.1	14.2	-1.5
Primary (noninterest) revenue and grants	31.6	30.8	30.6	28.8	29.2	29.3	29.3	29.3	29.4	175.2	
Primary (noninterest) expenditure	31.3	31.7	32.8	32.2	31.5	31.5	31.4	31.4	31.4	189.5	
Automatic debt dynamics [5/]	2.7	-0.8	2.1	-1.3	-0.9	-0.6	-0.7	-0.8	-0.9	-5.3	
Interest rate/growth differential [6/]	1.3	-1.6	-1.8	-1.3	-0.9	-0.6	-0.7	-0.8	-0.9	-5.3	
Of which: real interest rate	2.8	0.2	-1.1	-1.3	-0.1	0.3	0.2	0.2	0.2	-0.4	
Of which: real GDP growth	-1.5	-1.8	-0.7	-0.1	-0.8	-0.9	-0.9	-1.1	-1.1	-4.9	
Exchange rate depreciation [7/]	1.4	0.8	3.8	...	...	...	...	...	...	...	
Other identified debt-creating flows	0.4	1.0	-0.2	-0.5	-0.4	-0.7	-0.7	-0.6	-0.6	-3.4	
General Government: Net Privatization Proceeds (negative)	-0.5	-0.4	-0.2	-2.0	-1.2	-1.0	-1.0	-0.9	-0.9	-7.0	
Contingent liabilities	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	
General Government: Financing: Net Acquisition of Financial Assets	0.8	1.4	0.0	1.6	0.8	0.3	0.3	0.3	0.3	3.6	
Residual, including asset changes [8/]	-3.8	-1.1	-2.2	0.5	-0.1	0.0	0.5	0.6	0.5	1.9	



Source: IMF staff.

1/ Public sector is defined as General Government.

2/ Based on available data.

3/ Long-term bond spread over U.S. bonds.

4/ Defined as interest payments divided by debt stock (excluding guarantees) at the end of previous year.

5/ Derived as [(r - π(1+g) - g + ae(1+r)]/(1+g+π+gπ)) times previous period debt ratio, with r = interest rate; π = growth rate of GDP deflator; g = real GDP growth rate; a = share of foreign-currency denominated debt; and e = nominal exchange rate depreciation (measured by increase in local currency value of U.S. dollar).

6/ The real interest rate contribution is derived from the numerator in footnote 5 as r - π (1+g) and the real growth contribution as -g.

7/ The exchange rate contribution is derived from the numerator in footnote 5 as ae(1+r).

8/ Includes asset changes and interest revenues (if any). For projections, includes exchange rate changes during the projection period.

9/ Assumes that key variables (real GDP growth, real interest rate, and other identified debt-creating flows) remain at the level of the last projection year.

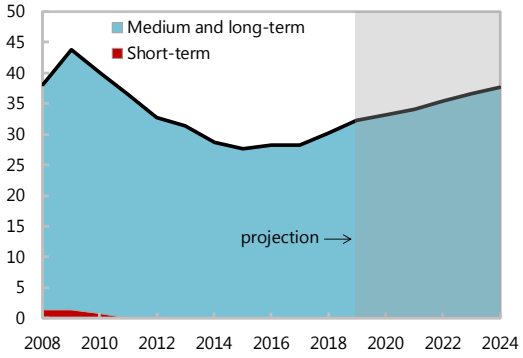
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Figure 2. Turkey Public DSA—Composition of Public Debt and Alternative Scenarios

Composition of Public Debt

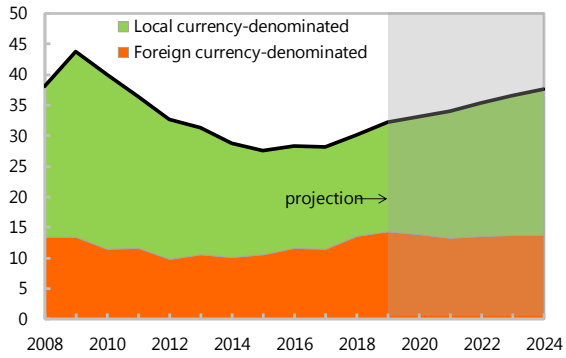
By Maturity

(in percent of GDP)

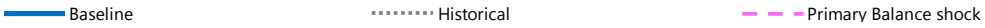


By Currency

(in percent of GDP)

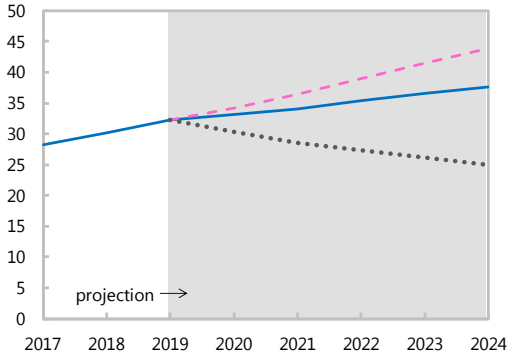


Alternative Scenarios



Gross Nominal Public Debt

(in percent of GDP)



Underlying Assumptions

(in percent)

Baseline Scenario	2019	2020	2021	2022	2023	2024
Real GDP growth	0.2	3.0	3.0	3.0	3.5	3.5
Inflation	14.4	11.8	11.6	10.9	10.8	11.0
Primary Balance	-3.4	-2.3	-2.2	-2.2	-2.1	-2.1
Effective interest rate	9.6	11.9	13.1	12.0	11.9	12.0
Primary Balance Shock						
Real GDP growth	0.2	3.0	3.0	3.0	3.5	3.5
Inflation	14.4	11.8	11.6	10.9	10.8	11.0
Primary Balance	-3.4	-2.9	-2.8	-2.2	-2.1	-2.1
Effective interest rate	9.6	11.9	13.2	12.2	12.1	12.1

Historical Scenario	2019	2020	2021	2022	2023	2024
Real GDP growth	0.2	5.3	5.3	5.3	5.3	5.3
Inflation	14.4	11.8	11.6	10.9	10.8	11.0
Primary Balance	-3.4	-0.1	-0.1	-0.1	-0.1	-0.1
Effective interest rate	9.6	11.9	13.2	12.2	12.1	12.1

Source: IMF staff.

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